

Late Radiation Failures after Iodine 125 Brachytherapy for Uveal Melanoma Compared with Charged-particle (Proton or Helium Ion) Therapy

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Objective: To evaluate late (more than 5 years) radiation failures after uveal melanoma treatment.

Design: Comparison of three retrospective, interventional, partially randomized case series.

Participants: Nine hundred ninety-six patients who were treated in several phase I, II, and III trials of uveal melanoma radiation.

Main Outcome Measures: Follow-up period, treatment history, recurrence rates, type of recurrence, and mortality associated with late local recurrences.

Results: Eleven of 996 irradiated uveal melanoma patients experienced intraocular recurrence more than 5 years after radiation. All 11 of these patients were treated with iodine 125 (¹²⁵I) brachytherapy. Late recurrences were detected between 5.5 to 15.3 years after treatment. These patients did not have either high-risk clinical parameters (thin, posterior tumors in proximity to the optic nerve) or radiation dosimetry characteristics (low dose-delivery radiation) associated with a known increased risk for tumor recurrence after radioactive plaques. The annualized incidence rate for regrowth was 1.9% per year between 5 and 15 years after ¹²⁵I brachytherapy. In contrast to charged particles, the risk of late recurrence after ¹²⁵I brachytherapy continued with increased follow-up.

Conclusions: There was a significantly higher late recurrence rate with ¹²⁵I brachytherapy as compared with charged particle radiation. Although tumor enlargement 5 or more years after radiation can be the result of intratumor hemorrhage, in a patient treated with radioactive plaque, a late failure is a distinct possibility. *Ophthalmology* 2002;109:1850–1854 © 2002 by the American Academy of Ophthalmology.

Radiation therapy for uveal melanoma has been used since 1929.¹ Several authors have documented retrospectively uveal melanoma radiation control rates between 60% and 97%, with various forms of brachytherapy and charged particles.^{2–15} In a prospective, randomized study, we observed significantly better local control with charged particles as compared with iodine 125 (¹²⁵I) brachytherapy.¹⁶

The patterns of treatment failures between charged particles and brachytherapy have been disparate. Local intraocular tumor failures more than 5 years after charged particle radiation are rare.^{16,17} In contrast, a number of isolated reports document radioactive plaque failures as long as 20 years after treatment.^{7,8} In this report, we describe 11 uveal melanoma patients who appeared to have good long-term tumor control after ¹²⁵I brachytherapy and in whom local

tumor recurrence then developed between 5.5 and 16 years after treatment.

Materials and Methods

We retrospectively reviewed all uveal melanoma patients we treated with helium ions (n = 348), protons (n = 199), or ¹²⁵I brachytherapy (n = 449) between 1978 and 2000 to assess the occurrence rate of late failures, defined as those detected at least 5 years after treatment. All patients were evaluated and treated in a standardized manner and were part of ongoing phase I, II, and III treatment trials. The surgical, radiation planning, and follow-up protocols were identical throughout the period of these studies. All tumors were localized with point source transillumination and indirect ophthalmoscopy as well as with diffuse corneal transillumination as previously described.⁷ Radiation treatment planning was performed in the same manner throughout the study. All patients treated with ¹²⁵I brachytherapy received 70 Gy to 1 mm above the apex of the tumor with a horizontal margin of at least 1.5 mm. Patients treated with charged particles received 48, 50, 60, 70, or 80 GyE of radiation to the tumor and 1.5 to 2.0 mm surrounding the tumor. In a prior study, we demonstrated that there was no difference in control, complications, or survival rates at these different doses.⁴ Radiation with particles was delivered over a 5- to 7-day period in four or five sessions. Treatment with brachytherapy encompassed a 4-day treatment course throughout this entire period. Patients were examined four times in the first year,

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three in the second year, and twice in the third year, then yearly unless a problem developed. We defined the posttreatment intraocular regrowth as sufficient tumor enlargements to mandate further therapy (e.g., enucleation, repeat radiation, or laser therapy). All 11 patients met this criterion. They demonstrated unequivocal enlargement documented with serial photographs for horizontal enlargement, serial quantitative ultrasonography, or both to demonstrate increased thickness.

Follow-up was obtained from our ophthalmic oncology practice, from the referring doctors, or from family members. For patients who died, we reviewed their terminal illness with their attending physicians and, when possible, obtained autopsy data. Cumulative hazard plots were constructed. The radioactive plaque-treated patients were part of either phase II or phase III studies. In the latter group, a randomized, prospective, dynamically balanced study was performed.⁷ These patients were randomized between helium ion and ^{125}I brachytherapy. In the nonrandomized portion of this study, the plaque patients had slightly smaller tumors than those treated with particles, and their tumors were less likely to involve the anterior uvea.⁷ The overall length of follow-up was slightly longer in the charged particle group as compared with the brachytherapy group. There was no difference in the local failure rates between patients in the randomized study as compared with those in the nonrandomized study with protons, helium ions, or ^{125}I brachytherapy.

Statistical analysis was performed using counting process models.¹⁸ Annualized incidence rates were calculated based on the number of tumor recurrences per person per year of follow-up and were compared using the likelihood ratio test.¹⁸ Cumulative hazard and cumulative incidence plots were constructed.¹⁸ The slope of the cumulative hazard represents the rate of tumor recurrence. The cumulative incidence estimates the number of patients in whom tumor recurrence will develop.

Results

In irradiated uveal melanoma patients treated with brachytherapy, 58 of 449 melanomas recurred locally with a mean interval between treatment and detection of recurrent growth of 3.6 years; the median was 2.9 years (range, 0.2–15.3 years). In contrast, after charged-particle therapy, intraocular tumor recurrence developed in 22 of 547 patients (15 of whom received helium ions and 7 of whom were treated with protons). The mean duration between treatment and the detection of intraocular recurrence was 1.6 years; the median was 0.9 years (range, 0.1–5.0 years).

Eleven uveal melanoma patients treated with ^{125}I brachytherapy had late intraocular tumor recurrences detected from 5.5 to 15.3 years (median, 7.5 years) after treatment. Ten were diffuse recurrences with generalized enlargement. One patient had a recurrence at the edge of the treated melanoma. All of these patients had regular follow-up examinations by the authors. Only one of these cases would have been considered at higher than average risk for local recurrence, given our previous data.¹⁴ This latter melanoma was in a subfoveal location and measured $9 \times 8 \times 4.5$ mm, but was located more than 2 mm distal to the optic nerve. Prior publications have documented several factors that are associated with a higher risk of intraocular brachytherapy failure, including thin tumors contiguous to the disc, or those tumors treated with a low radiation dose rate.¹⁴ The mean pretreatment melanoma thickness was 4.6 mm (range, 3.1–6.1 mm). Five tumors were located nasal to the optic disc (the median distance of the posterior tumor edge to optic nerve was 4.3 mm; range, 1.5–7 mm). One tumor was equatorial and five were 2 to 5 mm lateral to the fovea.

The mean interval between detection of intraocular regrowth and the last follow-up in living patients was 4.5 years (range,

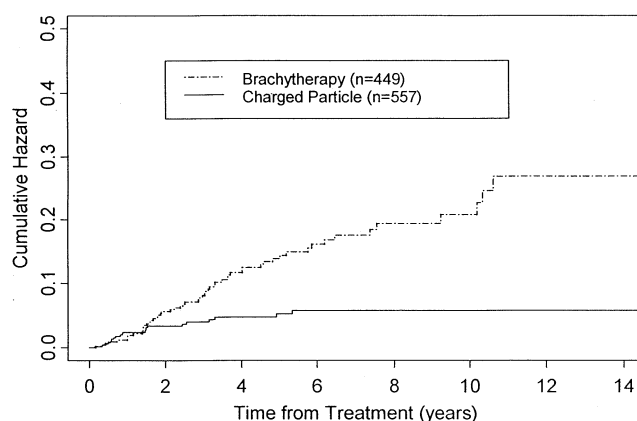


Figure 1. Cumulative hazard of local recurrence after radiation.

0.2–6 years.). Nine of 11 eyes were enucleated. One patient was re-treated with protons with a good intraocular tumor regression, but experienced metastases 3 years later and died. One patient's area of regrowth was at the edge of the irradiated melanoma and was treated with a laser, with excellent tumor control 7 years later. One patient had metastases detected within 6 months of recurrent intraocular tumor growth and enucleation; she is still living. An additional patient developed metastases 3 years later after the recurrence followed by enucleation and died 2 months later. All living patients have been contacted in 2000.

Histologic data were available in nine cases. Seven of nine patients had mixed cell melanomas and two were spindle cell tumors; all tumors had active melanoma cell cycling with between 0 and 15 mitoses per high-powered field. The one case with zero mitoses in the intraocular tumor had a contiguous, extraocular nodule with three mitoses per high-power field in that area of the tumor.

The annualized incidence rates of recurrent growth were significantly different in the charged-particle- versus the brachytherapy-treated uveal melanoma patients. Data from both the randomized, prospective ^{125}I versus the charged-particle treatment trials as well as the phase II study are congruent. Overall, the patients treated with charged particles had a longer follow-up and slightly larger tumors than those treated with brachytherapy.^{7,16} The recurrent growth rate after ^{125}I brachytherapy between 0 and 5 years was 2.8% per year and after 5 years was 1.9% per year ($P = 0.221$). In contrast, after charged particles it was 1.23% between 0 and 5 years after treatment and 0.09% per year 5 years after treatment ($P < 0.001$). This trend for continued risk of local recurrence in patients after ^{125}I brachytherapy is shown in Figure 1. Although the slope of the cumulative hazard (which represents the rate of local recurrence) flattens in patients after charged particle treatment, the risk of local recurrence in patients treated with ^{125}I brachytherapy has not significantly diminished through year 11.

The higher risk of local recurrence in patients after ^{125}I brachytherapy results in a significantly higher probability of local recurrence with increasing follow-up (Fig 2). For example, the 5- and 10-year cumulative incidence for local recurrence was 4.6% and 4.9%, respectively, after charged-particle treatment and 11.5% and 15.8%, respectively, after ^{125}I brachytherapy. In addition, the trend continued, and by year 12, the cumulative incidence was 18.9% for the brachytherapy group compared with 4.9% for the charged-particle group.

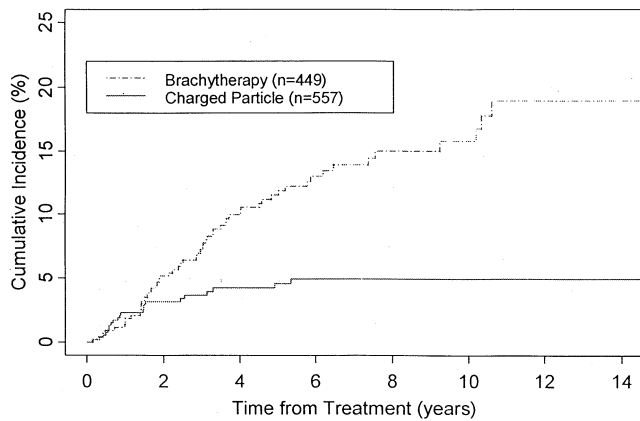


Figure 2. Probability of local recurrence after radiation.

Discussion

In our experience with charged-particle uveal melanoma radiation, the longest time between radiation and detection of intraocular melanoma recurrence is approximately 5 years, with a mean interval of 1.6 years.¹⁴ Similar data has been published from Gragoudas⁹ and Gragoudas et al¹⁷ in Boston; they observed one failure 10 years after proton radiation. In contrast, we report 11 cases of late intraocular radiation failures after ¹²⁵I brachytherapy between 5.5 to 15 years after treatment. The reasons for these late recurrences after radioactive plaque therapy are uncertain.

We first became concerned about delayed recurrence after brachytherapy several years ago when prostate brachytherapy showed some late recurrences. Ragde et al¹⁹ reported 219 prostate carcinoma patients irradiated before 1990; 25% of the failures occurred more than 5 years after radiation. For other sites, some late radiation failures have been reported, including nasopharyngeal carcinoma, cranio-pharyngioma, and skin cancers.^{20,21}

In several other ophthalmic oncologic reports, late failures after brachytherapy have been noted, but there is limited data. In two studies from Philadelphia, local tumor recurrence within 5 years was discussed.^{11–13} Vrabec et al¹² noted late failures after cobalt plaques, and in eight cases that were not used in their case-control study, the median interval to relapse was 3.6 years (range, 0.25–10.2 years). Other reports also have observed late failures, but the causes and differences between those and early failures were not discussed.^{8,10} Lommatzsch et al⁸ noted two local radiation failures occurring more than 20 years after treatment. In several uveal melanoma radiation studies, local failure after brachytherapy has been associated with increased tumor diameter, proximity to the optic nerve, increased tumor thickness, lower radiation dose rate, and a lower dose to the tumor apex.^{6–8,10–12,14,22,23} Our study was similar to the above reports in the mean interval to recurrence after brachytherapy, but we were unable to identify any differences in risk factors between those patients who relapsed earlier versus those who relapsed later than 5 years after radiation. As mentioned above, none of our late-failure patients would have been considered high risk for local brachytherapy failure using the above criteria.

There are several differences between uveal melanoma charged-particle teletherapy and ¹²⁵I brachytherapy. We delivered between 48- and 80-Gy equivalents of either protons or helium ions (physical gray $\times$ 1.2 [helium] or 1.1 [protons]). Either form of teletherapy delivers a uniform radiation dose to the tumor and adjacent sclera in four or five fractions over a 4- to 7-day period. In contrast, any form of brachytherapy delivers exponentially more radiation to the sclera and tumor base as compared with the tumor apex in a constant but slower delivery rate. To ensure a minimum tumor dose of 70 Gy with ¹²⁵I plaques, we have treated an additional margin of 1 to 2 mm horizontally and 1 mm vertically.²⁴ Some tumors, such as melanomas that are relatively large in diameter and thin, have had a higher failure rate, although none of our late failure cases would fit in that group.^{14,23} Similarly, we and others noted that tumors that abut the lateral edge of the optic nerve are also at higher risk for radioactive plaque failure, but none of our late failure cases were located in this area, although one was subfoveal.²³

Quivey et al²³ have discussed the effect of uveal melanoma brachytherapy dose rate; there was a significantly better tumor control with higher, as compared with lower, dose rates.²³ We doubt this later factor was operative in our late failures because we have always used a high ¹²⁵I dose rate.²⁴

It is conceivable that our late failures were the result of inadequate radiation dose delivery to these melanomas. We have reviewed all the failures with both clinical photographs and isodose curves and were not able to detect, with the margins used and the delineation of plaque position by diathermy spots, that any treatment delivered less than 70 Gy to the entire tumor. A number of tumors, given the geometry of the neoplasm, received significantly more radiation.

There have been a number of reports of errors in tumor localization and problems with plaque carrier placement on the scleral surface overlying an intraocular tumor.^{25–27} In a prior histopathologic study of patients treated with helium ions, we found that the surgical position of tantalum marker rings matched within 0.5 mm the position of rings in enucleated eyes. Our experience has been that placement of a radioactive plaque is technically easier than positioning rings for charged-particle radiation.²⁴ It is conceivable, despite several strategies we have used to prevent plaque-scleral surface misalignment, that physical placement issues compromised the radiation dose to some of these tumors. It is surprising, however, if this paradigm were likely, that only 1 of 10 cases had a marginal as opposed to a diffuse recurrence. If the later problem were operative, we would expect that most of the failures would be marginal misses, most likely on the posterior edge of the treated tumor, which we did not observe.

An alternative hypothesis for the lower efficacy of brachytherapy, as compared with charged particle radiation, could be a tumor bed effect. Clifton and Jirtle²⁸ hypothesized that the radiation tumor bed effect was the result of suppression of vascularization; several investigators have modeled this effect.^{29,30} Tumors with low clonogenic ability and low macrophage content, such as melanomas, are more likely to manifest this effect in a setting of inhomogeneous

radiation.^{31–33} In uveal melanomas in which the minimum radiation dose was inadequate to sterilize all viable tumor cells, the very high dose to the tumor base probably damaged the vasculature, selected more radio-resistant clones, and slowed the regrowth of surviving cells.^{31–35}

In this small study, we did not note as high a tumor-related mortality as we would have expected. Only two patients died of metastatic melanoma, and one additional patient is alive with disease outside of the eye. Several prior reports have documented increased tumor-associated mortality after local recurrence.^{12–14} We suspect that our failure to document that finding in this study is the result of the small number of cases. Alternatively, it is possible that these tumors, having a much slower growth rate, may have had a lower number of metastatic events associated with them. Although postrecurrence follow-up has been limited, the mean length of time that patients have been evaluated after a local failure was 4.5 years.

It is clear that reproductive integrity was not destroyed in all these melanoma cells after uveal melanoma brachytherapy, and as a result, a late failure occurred. All irradiated uveal melanoma patients require perpetual serial evaluations. Unlike patients who had charged particle radiation in which tumor enlargement more than 5 years later is much more likely to be the result of radiation-induced intratumor hemorrhage, in a patient treated with a radioactive plaque, a late failure is also a distinct possibility, although most intraocular tumor failures occur earlier.

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